Research Workflow

Stage-wise comparison of sentiment classification, complaint discovery and feedback exploration

DATA SOURCES

Amazon
product reviews

Yelp
business reviews

Twitter US Airline
sentiment

DATA PREPARATION

Schema unification · text cleaning · rating-derived labelling
class balancing · stratified 80/20 split (seed 42)

Unified corpus — 642,692 reviews
negative 218,799 · neutral 212,503 · positive 211,390

MODELLING

Stage 1
Sentiment classification

VADER (lexicon baseline)
Naive Bayes · Logistic regression
RoBERTa (pretrained)
DistilBERT (fine-tuned)

Stage 2
Complaint discovery

BERTopic per platform
121 raw topics
centroid merge → 24 categories
zero-shot assignment (τ = 0.35)

Stage 3
Feedback exploration

MiniLM embeddings (384-d)
FAISS index · MMR retrieval
grounded generation (Llama 3.1)
standalone LLM baseline

EVALUATION

Held-out set n = 128,539
accuracy · macro F1
per-class · per-platform
confusion matrices
McNemar exact test

No ground truth available
centroid separation
within-category coherence
coverage · platform spread
manual inspection

RAGAS, n = 5 questions
faithfulness
answer relevancy
context precision / recall
latency

Diagnostic experiments
per-platform · binary re-evaluation
VADER threshold sweep
balanced sample n = 1,800

Comparative findings
RQ1 method ranking is conditional on domain · RQ2 grounding versus responsiveness
RQ3 taxonomy coverage limits · RQ4 traceability of generated answers

OUTPUTS

All metrics written to data/results/ — every reported figure traceable to its producing script

Note: the fine-tuned DistilBERT classifier was developed against a separately generated 70/15/15 split.
Its clean held-out result is reported throughout; the limitation is discussed in Section 3.5.3.

Raw data · Preparation · Sentiment · Discovery · Retrieval / generation · Evaluation